

Topic 8: Hypothesis Testing: One-sample Tests

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What is a Hypothesis?

- A hypothesis is a claim (assertion) about a particular parameter of a population.
 - Population mean example: The mean monthly cell phone bill in this city is $\mu = \$42$.
 - Population proportion example: The proportion of adults in this city with cell phones is $\pi = 0.68$.



The Null Hypothesis, H_0 (1 of 2)

- States the claim or assertion to be tested (status quo).
 - Example: The mean diameter of a manufactured bolt is 30mm ($H_0 : \mu = 30$).
- Is always about a population parameter, not about a sample statistic.

$$H_0 : \mu = 30$$

$$\cancel{H_0 : \bar{X} = 30}$$



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The Null Hypothesis, H_0 (2 of 2)

- Begin with the assumption that the null hypothesis is true.
 - Similar to the notion of innocent until proven guilty.



- Refers to the status quo or historical value.
- Always contains “=”, “ $\leq$ ” or “ $\geq$ ” sign.
- May or may not be rejected.



The Alternative Hypothesis, H_1

- Is the opposite of the null hypothesis.
 - e.g. The mean diameter of a manufactured bolt is not equal to 30mm ($H_1 : \mu \neq 30$).
- Challenges the status quo.
- Can only contain either the “<”, “>” or “ $\neq$ ” sign.
- May or may not be proven.
- Is generally the hypothesis that the researcher is trying to prove.

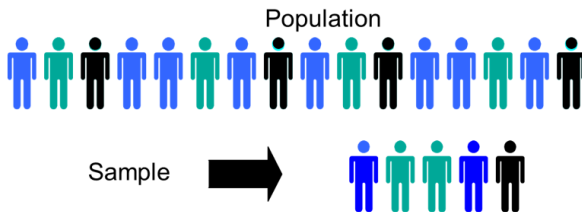


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The Hypothesis Testing Process (1 of 3)

- Claim: The population mean age is 50.
 - $H_0 : \mu = 50$, $H_1 : \mu \neq 50$
- Sample the population and find the sample mean.



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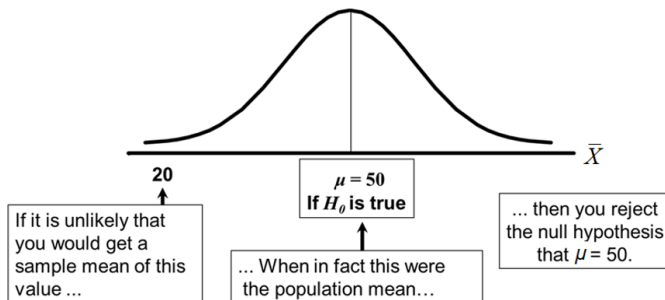
The Hypothesis Testing Process (2 of 3)

- Suppose the sample mean age was $\bar{X} = 20$.
- This is significantly lower than the claimed mean population age of 50.
- If the null hypothesis were true, the probability of getting such a different sample mean would be very small, so you reject the null hypothesis.
- In other words, getting a sample mean of 20 is so unlikely if the population mean was 50, you conclude that the population mean must not be 50.



The Hypothesis Testing Process (3 of 3)

Sampling distribution of $\bar{X}$



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The Test Statistic and Critical Values (1 of 3)

- If the sample mean is close to the stated population mean, the null hypothesis is not rejected.
- If the sample mean is far from the stated population mean, the null hypothesis is rejected.
- How far is 'far enough' to reject H_0 ?



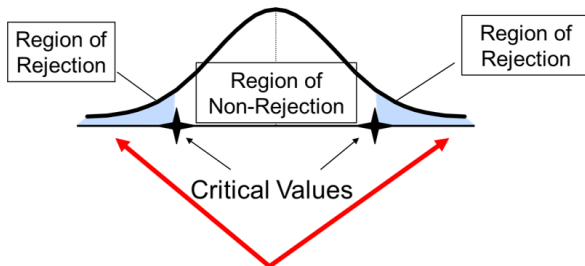
The Test Statistic and Critical Values (2 of 3)

- The **test statistic** is a value derived from ample data that is used to determine whether the null hypothesis should be rejected or not.
- The sampling distribution of the test statistic is divided into two regions, a **region of rejection** and a **region of non-rejection**.
- The critical value of a test statistic creates a “line in the sand” for decision making – it answers the question of how far is far enough.
- To make a decision concerning the null hypothesis, you first determine the **critical value** of the test statistic.



The Test Statistic and Critical Values (3 of 3)

Sampling distribution of the test statistic



“Too Far Away” From Mean of Sampling Distribution



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Risks in Decision Making Using Hypothesis Testing

- **Type I error**

- Reject a true null hypothesis.
- A type I error is a “false alarm”.
- The probability of a Type I error is α .
 - ▷ called level of significance of the test
 - ▷ set by researcher in advance

- **Type II error**

- Fail to reject a false null hypothesis.
- Type II error represents a “missed opportunity”.
- The probability of a Type II error is β .



The Level of Significance, α

- Defines the unlikely values of the sample statistic if the null hypothesis is true.
 - Defines **rejection region** of the sampling distribution.
- Is designed by α (level of significance).
 - Typical values are 0.01, 0.05 or 0.10.
 - Note relationship to 99%, 95% and 90% confidence levels.
- Is selected by the researcher at the beginning.
- Provides the critical value(s) of the test.



Possible Errors in Hypothesis Test Decision Making (1 of 3)

Possible Hypothesis Test Outcomes		
	Actual Situation	
Decision	H_0 True	H_0 False
Do Not Reject H_0	No Error Probability $1 - \alpha$	Type II Error Probability β
Reject H_0	Type I Error Probability α	No Error Power $1 - \beta$



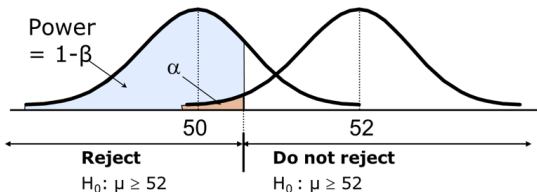
Possible Errors in Hypothesis Test Decision Making (2 of 3)

- The **confidence coefficient**, $1 - \alpha$, is the probability of not rejecting H_0 , when it is true and should not be rejected.
- The confidence level of a hypothesis test is: $(1 - \alpha) \times 100\%$.



Possible Errors in Hypothesis Test Decision Making (3 of 3)

- The **power of a statistical test**, $1 - \beta$, is the probability that you will reject the null hypothesis when it is false and should be rejected.
 - Suppose we correctly reject $H_0 : \mu \geq 52$ when in fact the true mean is $\mu = 50$.



- A one-tail test is more powerful than a two-tail test.



Type I & II Error Relationship

- Type I and Type II errors cannot happen at the same time.
 - A Type I error can only occur if H_0 is true.
 - A Type II error can only occur if H_0 is false.

If Type I error probability (α) $\uparrow$, then Type II error probability (β) $\downarrow$

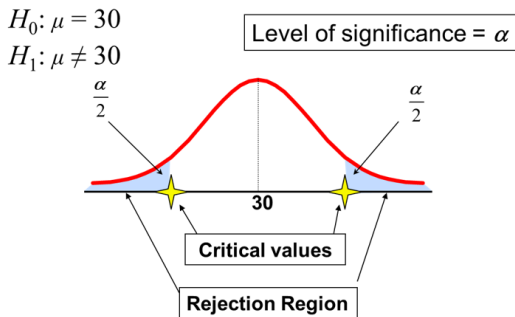


Factors Affecting Type II Error

- All else equal,
 - $\beta \uparrow$ when the difference between hypothesized parameter and its true value $\downarrow$
 - $\beta \uparrow$ when $\alpha \downarrow$
 - $\beta \uparrow$ when $\sigma \uparrow$
 - $\beta \uparrow$ when $n \downarrow$



Level of Significance and the Rejection Region



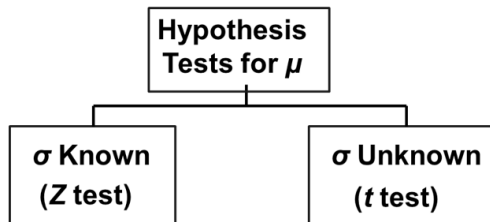
This is a two-tail test because there is a rejection region in both tails.



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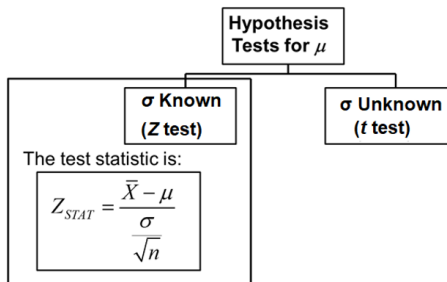


Hypothesis Tests for the Mean



Z Test of Hypothesis for the Mean (σ is known)

- The Z test of hypothesis for the mean is a test about the population mean which uses the standard normal distribution.



Critical Value Approach to Testing

- For a two-tail test for the mean, σ is known:
 - Convert sample statistic ($\bar{X}$) to the test statistic (Z statistic).
 - Determine the critical Z values for a specified level of significance α by using Excel.
 - **Decision rule:** If the test statistic falls in the rejection region, reject H_0 ; otherwise do not reject H_0 .

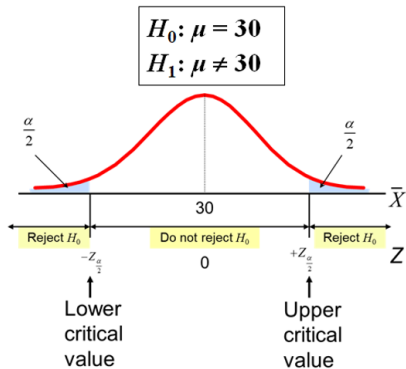


Two-Tail Tests

- A **two-tail test** is a hypothesis test where the rejection region is divided into the two tails of the probability distribution.
- There are two cutoff values (**critical values**), defining the regions of rejection.



Illustration



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6 Steps in Hypothesis Testing (1 of 2)

- 1 State the null hypothesis, H_0 , and the alternative hypothesis, H_1 .
- 2 Choose the level of significance, α , and the sample size, n .
 - The level of significance is based on the relative importance of Type I and Type II errors.
- 3 Determine the appropriate test statistic and sampling distribution.
- 4 Determine the critical values that divide the rejection and non-rejection regions.



6 Steps in Hypothesis Testing (2 of 2)

- ⑤ Collect data and calculate the value of the test statistic.
- ⑥ Make the statistical decision and state the managerial conclusion.
 - If the test statistic falls into the non-rejection region, do not reject the null hypothesis H_0 ; if the test statistic falls into the rejection region, reject the null hypothesis.
 - Express the managerial conclusion in the context of the real-world problem.



Hypothesis Testing Example (1 of 4)

Test the claim that the true mean diameter of a manufactured bolt is 30mm. (assume $\sigma = 0.8$)

Solution:

1. State the appropriate null and alternative hypotheses:
 - $H_0 : \mu = 30$, $H_1 : \mu \neq 30$ (This is a two-tail test)
2. Specify the desired level of significance and the sample size:
 - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test.



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Hypothesis Testing Example (2 of 4)

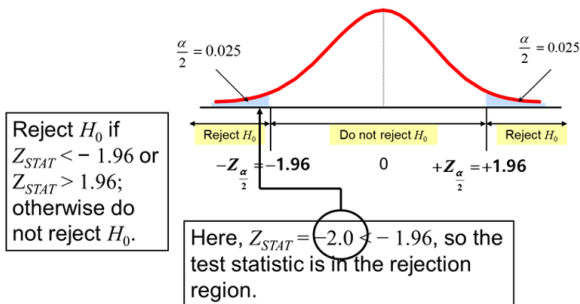
3. Determine the appropriate technique:
 - σ is assumed known so this is a Z test.
4. Determine the critical values:
 - For $\alpha = 0.05$ the critical Z values are ± 1.96 .
5. Collect the data and compute the test statistic:
 - Suppose the sample results are
 $n = 100$, $\bar{X} = 29.84$ ($\sigma = 0.8$ is assumed known)
So the test statistic is:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{29.84 - 30}{\frac{0.8}{\sqrt{100}}} = \frac{-0.16}{0.08} = -2.0$$



Hypothesis Testing Example (3 of 4)

6. Is the test statistic in the rejection region?

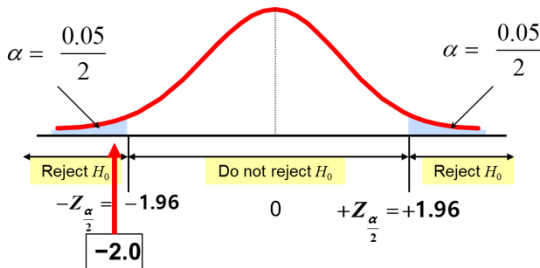


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Hypothesis Testing Example (4 of 4)

6. (continued). Reach a decision and interpret the result.



Since $Z = -2.0 < -1.96$, reject the null hypothesis and conclude there is sufficient evidence that the mean diameter of a manufactured bolt is not equal to 30mm.



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The p -value Approach to Testing

- The **p-value** is the probability of getting a test statistic more extreme than the sample results, given that the null hypothesis, H_0 , is true.
 - Also called *observed level of significance*.
 - Smallest value of α for which H_0 can be rejected.



Interpreting the p -value

- Obtain the p -value from Excel and compare it with α .
 - If $p\text{-value} < \alpha$, reject H_0 .
 - If $p\text{-value} \geq \alpha$, do not reject H_0 .
- Remember
 - If the p -value is low then H_0 must go.



The 5 Step p -value Approach to Hypothesis Testing

- 1 State the null hypothesis, H_0 and the alternative hypothesis, H_1 .
- 2 Choose the level of significance, α , and the sample size, n .
 - The level of significance is based on the relative importance of Type I and Type II errors.
- 3 Determine the appropriate test statistic and sampling distribution.
- 4 Collect data and compute the value of the test statistic and the p -value.
- 5 Make the statistical decision and state the managerial conclusion.
 - If the p -value is $< \alpha$ then reject H_0 ; otherwise do not reject H_0 .
 - State the managerial conclusion in the context of the real-world problem.



p -value Hypothesis Testing Example (1 of 4)

Test the claim that the true mean diameter of a manufactured bolt is 30mm. (assume $\sigma = 0.8$)

Solution:

1. State the appropriate null and alternative hypotheses:
 - $H_0 : \mu = 30$, $H_1 : \mu \neq 30$ (This is a two-tail test)
2. Specify the desired level of significance and the sample size:
 - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test.



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p-value Hypothesis Testing Example (2 of 4)

3. Determine the appropriate technique:
 - σ is assumed known so this is a Z test.
 4. Collect the data and compute the test statistic:
 - Suppose the sample results are
 $n = 100$, $\bar{X} = 29.84$ ($\sigma = 0.8$ is assumed known)
- So the test statistic is:

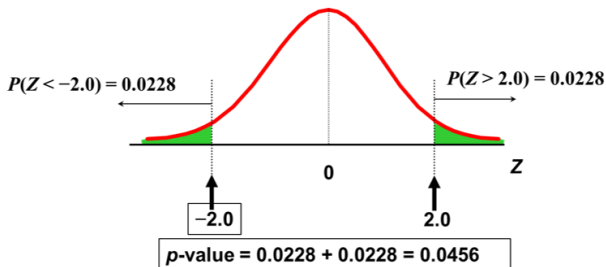
$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{29.84 - 30}{\frac{0.8}{\sqrt{100}}} = \frac{-0.16}{0.08} = -2.0$$



p -value Hypothesis Testing Example (3 of 4)

4. (continued) Calculate the p -value:

- How likely is it to get a Z of -2 (or something further from the mean (0), in either direction) if H_0 is true?



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p -value Hypothesis Testing Example (4 of 4)

5. Is the p -value $< \alpha$?
 - Since $p\text{-value} = 0.0456 < \alpha = 0.05$, therefore reject H_0 .
5. (continued) State the managerial conclusion in the context of the situation.
 - There is sufficient evidence to conclude the mean diameter of a manufactured bolt is not equal to 30mm.



You Do It

When asked to explain the meaning of “the p-value was $p = 0.03$ ”, a student says, “This means there is only probability 0.03 that the null hypothesis is true.” Is this an essentially correct explanation? Explain your answer.



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Connection Between Two Tail Tests and Confidence Intervals (1 of 2)

- For $\bar{X} = 29.84$, $\sigma = 0.8$ and $n = 100$, the 95% confidence interval is:

$$29.84 - (1.96)\frac{0.8}{\sqrt{100}} \text{ to } 29.84 + (1.96)\frac{0.8}{\sqrt{100}}$$
$$29.6832 \leq \mu \leq 29.9968$$

- Since this interval does not contain the hypothesized mean (30), we reject the null hypothesis at $\alpha = 0.05$.



Connection Between Two Tail Tests and Confidence Intervals (2 of 2)

- Although confidence interval estimation and hypothesis testing are based on the same set of concepts, they are used for different purposes.
- **Confidence intervals:** used to estimate parameters.
- **Hypothesis testing:** used for making decisions about specified values of population parameters.



Do You Ever Truly Know σ ?

- Probably not!
- In virtually all real world business situations, σ is not known.
- If there is a situation where σ is known then μ is also known (since to calculate σ you need to know μ).
- If you truly know μ there would be no need to gather a sample to estimate it.



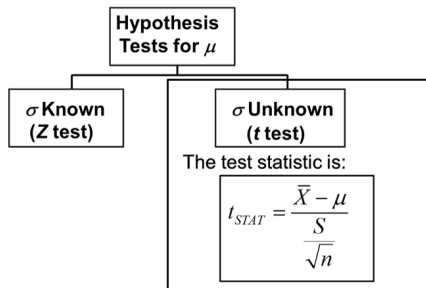
Hypothesis Testing: σ unknown

- If the population standard deviation is unknown, you instead use the sample standard deviation S .
- Because of this change, you use the t distribution instead of the Z distribution to test the null hypothesis about the mean.
- When using the t distribution you must assume the population you are sampling from follows a normal distribution.
- All other steps, concepts, and conclusions are the same.



t Test of Hypothesis for the Mean (σ unknown)

- The **t test of hypothesis for the mean** is a test about the population mean that uses a t distribution.



Example: Two-Tail Test (σ unknown)

The average cost of a hotel room in Perth is said to be \$168 per night. To determine if this is true, a random sample of 25 hotels is taken and resulted in an $\bar{X}$ of \$172.50 and an S of \$15.40. Test the appropriate hypotheses at $\alpha = 0.05$.



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Solution (1 of 2)

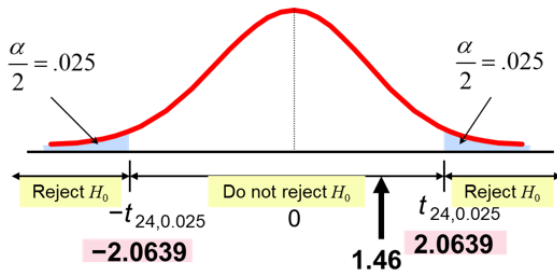
- $H_0 : \mu = 168, H_1 : \mu \neq 168$
- $\alpha = 0.05, n = 25, df = 25 - 1 = 24$
- σ is unknown, so use a t statistic:

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{172.50 - 168}{\frac{15.40}{\sqrt{25}}} = 1.46$$

- Critical value: $\pm t_{24,0.025} = \pm 2.0639$
- Do not reject H_0 : insufficient evidence that true mean cost is different from \$168



Solution (2 of 2)



To Use the t Test Must Assume the Population Is Normal

- As long as the sample size is not very small and the population is not very skewed, the t -test can be used.
- To evaluate the normality assumption:
 - Determine how closely sample statistics match the normal distribution's theoretical properties.
 - Construct a histogram or stem-and-leaf display or boxplot or a normal probability plot.



Two-Tail t Test Using A p -value from Excel

- Since this is a t -test we cannot calculate the p -value without some calculation aid.
- The Excel output below does this:

t Test for the Hypothesis of the Mean

Data		
Null Hypothesis $\mu=$	\$	168.00
Level of Significance		0.05
Sample Size		25
Sample Mean	\$	172.50
Sample Standard Deviation	\$	15.40

Intermediate Calculations

Standard Error of the Mean	\$	3.08	=B8/SQRT(B6)
Degrees of Freedom		24	=B6-1
t test statistic		1.46	= (B7-B4)/B11

Two-Tail Test

Lower Critical Value	-2.0639	=TINV(B5,B12)
Upper Critical Value	2.0639	=TINV(B5,B12)
p-value	0.157	=TDIST(ABS(B13),B12,2)
Do Not Reject Null Hypothesis		=IF(B18<B5, "Reject null hypothesis", "Do not reject null hypothesis")

$p\text{-value} > \alpha$
So do not reject H_0



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Connection of Two Tail t Test to Confidence Intervals

- For $\bar{X} = 172.5$, $S = 15.40$ and $n = 25$, the 95% confidence interval for μ is:

$$172.5 - (2.0639) \frac{15.4}{\sqrt{25}} \text{ to } 172.5 + (2.0639) \frac{15.4}{\sqrt{25}}$$
$$166.14 \leq \mu \leq 178.86$$

- Since this interval contains the hypothesized mean (168), we do not reject the null hypothesis at $\alpha = 0.05$.



One-Tail Tests

- In many cases, the alternative hypothesis focuses on a particular direction.

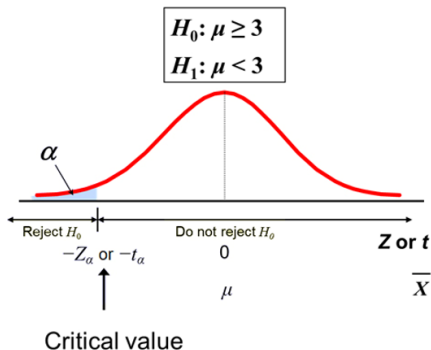
$H_0 : \mu \geq 3$
 $H_1 : \mu < 3$ $\Rightarrow$ This is a lower-tail test since the alternative hypothesis is focused on the lower tail below the mean of 3

$H_0 : \mu \leq 3$
 $H_1 : \mu > 3$ $\Rightarrow$ This is an upper-tail test since the alternative hypothesis is focused on the upper tail above the mean of 3



Lower-Tail Tests

- There is only one critical value, since the rejection area is in only one tail.

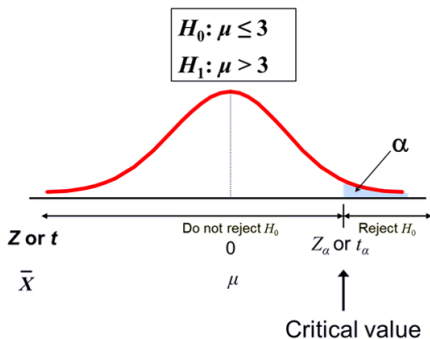


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Upper-Tail Tests

- There is only one critical value, since the rejection area is in only one tail.



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Example: Upper-Tail Test

A phone industry manager thinks that customer monthly cell phone bills have increased, and now average over \$52 per month. The company wishes to test this claim. (Assume a normal population)

Form hypothesis test:

$H_0 : \mu \leq 52$ the mean is not over \$52 per month

$H_1 : \mu > 52$ the mean **is** greater than \$52 per month
(i.e., sufficient evidence exists to support the manager's claim)

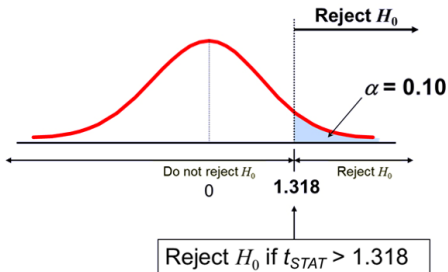


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Solution (1 of 4)

- Suppose that $\alpha = 0.10$ is chosen for this test and $n = 25$.
- Find the rejection region:



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Solution (2 of 4)

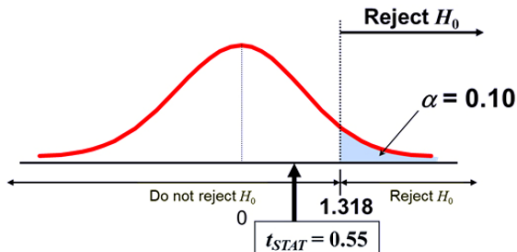
- Obtain sample and compute the test statistic.
- Suppose a sample is taken with the following results: $n = 25$, $\bar{X} = 53.1$ and $S = 10$.
 - Then the test statistic is:

$$t = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{53.1 - 52}{\frac{10}{\sqrt{25}}} = \boxed{0.55}$$



Solution (3 of 4)

- Reach a decision and interpret the result:

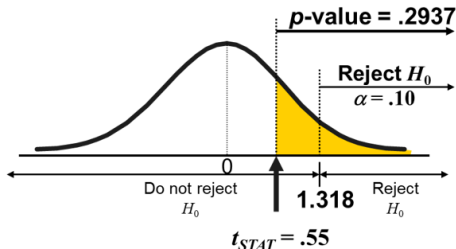


Do not reject H_0 since $t = 0.55 < 1.318$, there is not sufficient evidence that the mean bill is over \$52.



Solution (4 of 4)

- Calculate the p -value and compare to α (p -value below calculated using Excel spreadsheet on next page)



Do not reject H_0 since $p\text{-value} = 0.2937 > \alpha = 0.10$.



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Calculating the p -Value for the Upper Tail t Test Using Excel

t Test for the Hypothesis of the Mean

Data	
Null Hypothesis $\mu =$	52.00
Level of Significance	0.1
Sample Size	25
Sample Mean	53.10
Sample Standard Deviation	10.00

Intermediate Calculations	
Standard Error of the Mean	2.00
Degrees of Freedom	24
t test statistic	0.55

=B8/SQRT(B6)

=B6-1

=(B7-B4)/B11

Upper Tail Test	
Upper Critical Value	1.318
p-value	0.2937
Do Not Reject Null Hypothesis	

=TINV(2*B5,B12)

=TDIST(ABS(B13),B12,1)

=IF(B18<B5, "Reject null hypothesis",
"Do not reject null hypothesis")



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Hypothesis Tests for Proportions (1 of 2)

- Involves categorical variables.
- Two possible outcomes:
 - Possesses characteristic of interest
 - Does not possess characteristic of interest
- Fraction or proportion of the population in the category of interest is denoted by π .



Proportions

- Sample proportion in the category of interest is denoted by p

$$p = \frac{X}{n} = \frac{\text{number in category of interest in sample}}{\text{sample size}}$$

- When both $n\pi$ and $n(1 - \pi)$ are at least 5, p can be approximated by a normal distribution with mean and standard deviation:

$$\mu_p = \pi \quad \sigma_p = \sqrt{\frac{\pi(1 - \pi)}{n}}$$



Hypothesis Tests for Proportions (2 of 2)

- The sampling distribution of p is approximately normal, so the test statistic is a Z_{STAT} value:

$$Z_{STAT} = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Hypothesis Tests for p

$n\pi \geq 5$
and
 $n(1-\pi) \geq 5$

$n\pi < 5$
or
 $n(1-\pi) < 5$

Not discussed
in this chapter



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Z Test for Proportion in Terms of Number in Category of Interest

- An equivalent form to the last slide, but in terms of the number in the category of interest, X :

$$Z_{STAT} = \frac{X - n\pi}{\sqrt{n\pi(1-\pi)}}$$

Hypothesis Tests for X

$X \geq 5$
and
 $n - X \geq 5$

$X < 5$
or
 $n - X < 5$

Not discussed
in this chapter



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Example: Z Test for Proportion

A marketing company claims that it receives 8% responses from its mailing. To test this claim, a random sample of 500 were surveyed with 25 responses. Test at the $\alpha = 0.05$ significance level.



Check:

$$n\pi = 500 \times 0.08 = 40 > 5$$

$$n(1 - \pi) = 500 \times 0.92 = 460 > 5$$

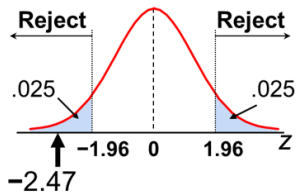


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Solution (1 of 3)

- $H_0 : \pi = 0.08, H_1 : \pi \neq 0.08$
- $\alpha = 0.05, n = 500, p = 0.05$
- Critical values: ± 1.96



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Solution (2 of 3)

- Test statistic:

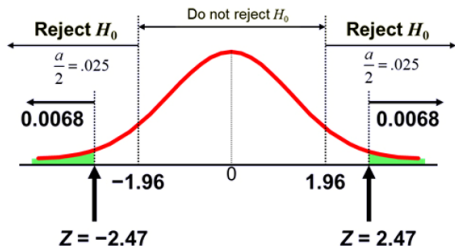
$$Z = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.05 - 0.08}{\sqrt{\frac{0.08(1-0.08)}{500}}} = \boxed{-2.47}$$

- Decision: Reject H_0 at $\alpha = 0.05$
- Conclusion: There is sufficient evidence to reject the company's claim of 8% response rate.



Solution (3 of 3)

- Calculate the p -value and compare to α (For a two-tail test the p -value is always two-tail.)



p -value = 0.0136:

$$\begin{aligned} P(Z \leq -2.47) + P(Z \geq 2.47) \\ = 2(0.0068) = \boxed{0.0136} \end{aligned}$$

- Decision: Reject H_0 since p -value = 0.0136 < $\alpha = 0.05$.



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Statistical Significance vs. Practical Significance

- Statistically significant results (rejecting the null hypothesis) are not always of practical significance
 - This is more likely to happen when the sample size gets very large.
- Practically important results might be found to be statistically insignificant (failing to reject the null hypothesis)
 - This is more likely to happen when the sample size is relatively small.



Potential Hypothesis Testing Pitfalls and Ethical Issues

- Use randomly collected data to reduce selection biases or coverage errors.
- Do not use human subjects without informed consent.
- Choose the level of significance, α , and the type of test (one-tail or two-tail) before data collection.
- Do not employ 'data snooping' to choose between one-tail and two-tail test, or to determine the level of significance.
- Do not practice 'data cleansing' to hide observations that do not support a stated hypothesis.
- Report all pertinent findings.
- Distinguish between statistical significance vs. practical significance.

